

AUTOMATIC PUBLIC SPEAKING QUALITY ANALYSIS

Predicting TED Talk Engagement Using Multi-Modal Machine Learning Techniques

PRESENTATION OVERVIEW

- Introduction to the Problem
- Timeline & Methodology
- Experimental Pipeline
- Results Overview
- Limitations
- Future Work
- Reflection

INTRODUCTION TO THE PROBLEM

CONTEXT

Communication

Public speaking is a critical communication skill.

Problem

Evaluating speech quality is subjective & costly
Expert annotation is difficult to obtain at scale.

Research Question

Can multi-modal machine learning predict public speaking engagement at scale?

MOTIVATION

Benefits

Understanding communication effectiveness.
Training tools for public speakers.

Machine Learning

An unbiased and scalable method to evaluate public speaking quality.

TED Talks

Offers a rich, standardised dataset including transcripts, audio, video and metadata
Enables building predictive and generalisable models

DATA COLLECTION



FEATURE EXTRACTION



MODEL TRAINING



EVALUATION

Construction of two separate but symmetrical TED talk datasets – one for text, one for audio



Transforming talks into numerical feature representations



Training and tuning several machine learning models on extracted feature sets



Comparison of predictive performance metrics between experiments



PIPELINE OVERVIEW

DATASET & TARGET VARIABLE



TED TALKS

Full speech texts
Raw audio recordings
Metadata



MODALITIES

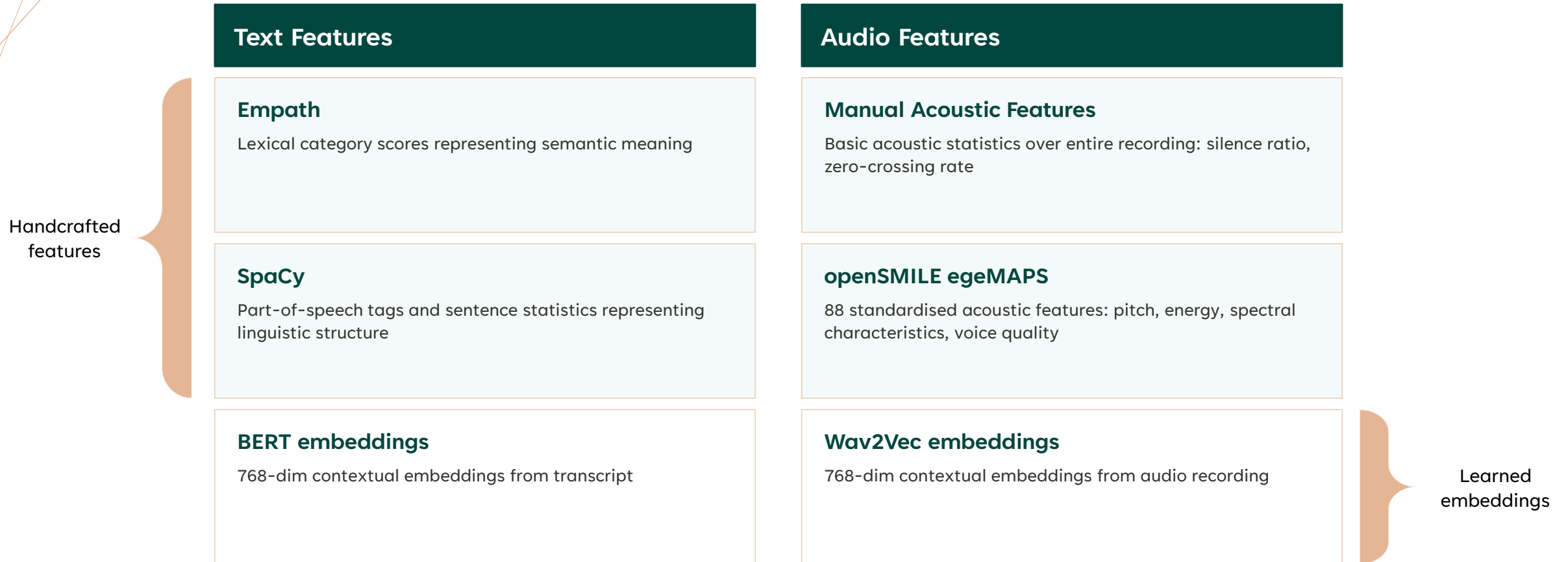
Linguistic features
Acoustic features



TARGET VARIABLES

log_views
log_views_per_day

FEATURE TYPES – MULTIMODAL APPROACH



EXAMPLE EXTRACTIONS FROM DATASET

TED Talk: How Healthy Living Nearly Killed Me – AJ Jacobs

Transcript: “I've spent the last decade subjecting myself to pain and humiliation, hopefully for a good cause, which is self-improvement. And I've done this in three parts. So first I started with the mind. And I decided to try to get smarter by reading...”

Empath

joy : 0.001641 , **communication** : 0.014766 , **business** : 0.002461 ... 14 feature categories

Manual Acoustic Features

silence_ratio : 0.121645 , **zcr_mean** : 0.133689 , **zcr_std** : 0.089967 ... 35 feature categories

SpaCy

char_count : 6633 , **noun_count** : 290 , **verb_count** : 712 ... 58 feature categories

openSMILE eGeMAPS

loudness_sma3_amean : 0.967142, **spectral_flux_amean** : 0.031752 ... 88 feature categories

BERT Embeddings

[-0.00991958, 0.09567304, 0.13948345 ...] 768 dimensions

Wav2Vec embeddings

[-0.034801, 0.006326, 0.208052 ...] 768 dimensions

EXPERIMENTAL SETUP

Models Evaluated

Linear Regression

Ridge Regression

Random Forest

Extra Trees

XGBoost

Support Vector Regressor (SVR)

Evaluation Metrics

RMSE

Root Mean Squared Error – overall prediction error magnitude

MAE

Mean Absolute Error – average absolute prediction error

Pearson r

Correlation coefficient – linear association between predicted and actual values

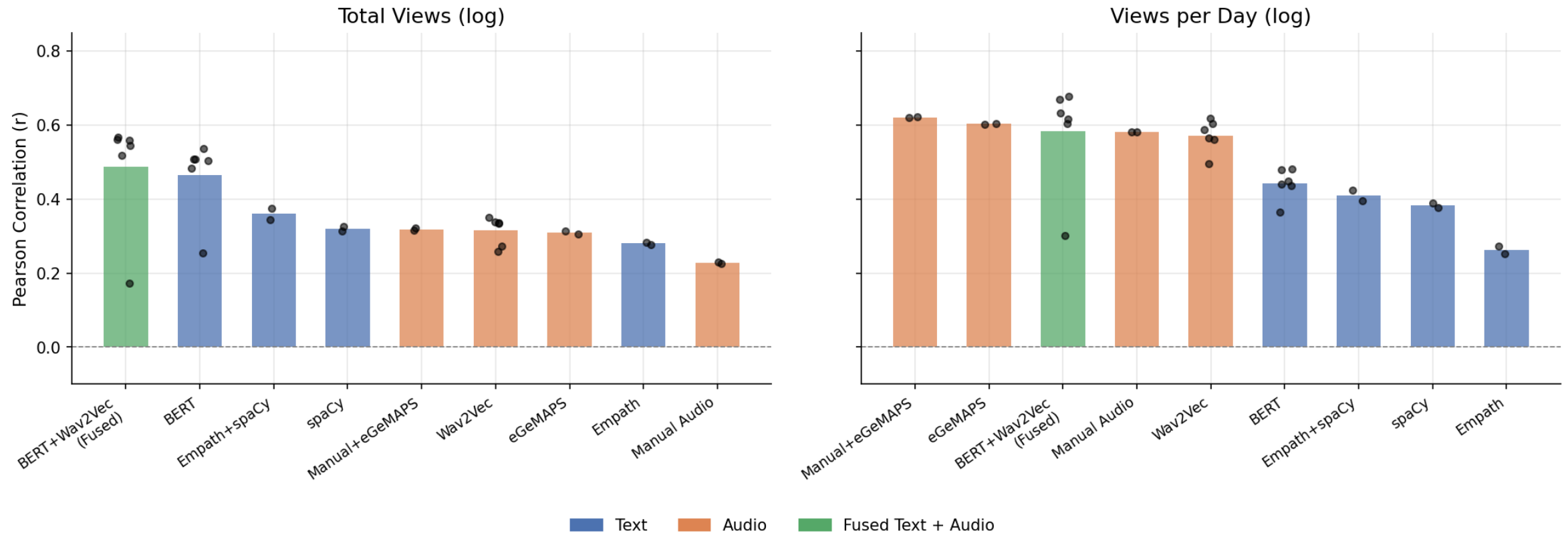


EXPERIMENTAL RESULTS (KEY MODELS) – LOG_VIEWS_PER_DAY

Feature Set	Best Model	RMSE	MAE	Pearson corr
BERT embeddings (text)	Ridge	0.929	0.751	0.482
Wav2Vec embeddings (audio)	SVR	0.821	0.628	0.619
Fused Bert + Wav2Vec embeddings (Multimodal)	Ridge	0.771	0.598	0.677

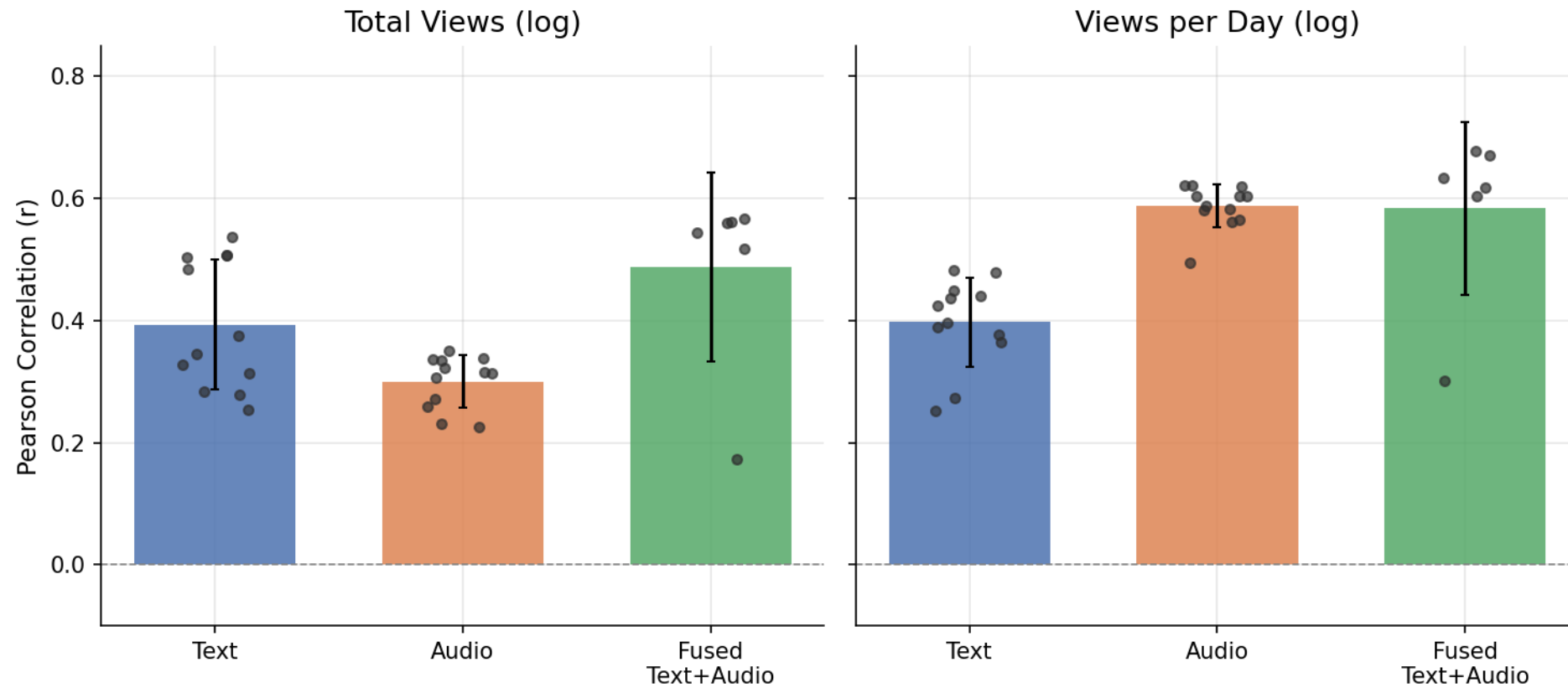
EXPERIMENTAL RESULTS

Figure 1 — Pearson Correlation by Feature Set
(bars = mean across models; dots = individual models)



EXPERIMENTAL RESULTS

Figure 2 — Correlation by Modality
(bars = mean; dots = individual featureset-model combinations)



WHAT DO THESE RESULTS TELL US?

Feature representation matters

Learned embeddings outperform handcrafted features

BERT and Wav2Vec capture richer semantic and acoustic meaning

Modality differences are clear

Audio features achieve highest correlation ($r \approx 0.62$)

Suggests *how something is said* strongly influences engagement

Fusion improves prediction error

Combined model achieves lowest RMSE (0.585)

Indicates complementary information between text and audio

Trade-off between metrics

Fusion improves overall error

But does not outperform audio-only on correlation

EXAMPLE PREDICTION FROM TEST SET

TED Talk: How Healthy Living Nearly Killed Me – AJ Jacobs

Transcript: “I've spent the last decade subjecting myself to pain and humiliation, hopefully for a good cause, which is self-improvement. And I've done this in three parts. So first I started with the mind. And I decided to try to get smarter by reading...”

Model Used

Fused Audio and Text Feature Embeddings
SVR Model

Predicted log_views:

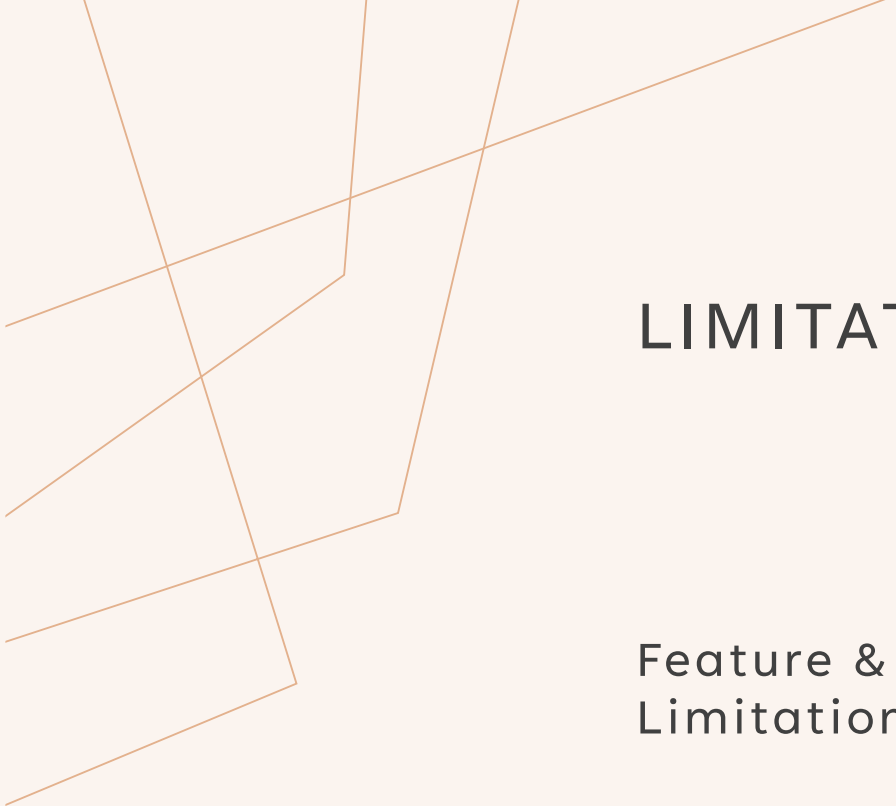
14.205

Actual log_views:

14.200

Absolute Error:

0.005



LIMITATIONS

Feature & Modelling Limitations

Visual cues (gestures, facial expressions) not captured

Deep models require more data to achieve strong generalization

Limited temporal modelling

Dataset Limitations

Engagement $\neq$ true quality

Influenced by: Topic popularity, speaker fame, promotion, platform growth

FUTURE WORK

Add Visual Modality

Incorporate facial expressions and speaker gestures as additional feature streams to capture non-verbal communication cues.

Larger & More Diverse Datasets

Extend to YouTube talks and academic conference presentations for broader coverage, scale, and speaker diversity.

Deep Multimodal Models

Explore other transformer-based architectures for joint text and audio modelling

Practical Applications

Develop speaker coaching tools and recommendation systems based on the trained pipeline to provide actionable feedback.

REFLECTION

What Worked Well	Key Challenges	What I'd Do Differently
Structured modular pipeline was effective and reproducible Controlled multimodal experiments produced clear, comparable results Validation-based hyperparameter tuning improved reliability	Feature alignment across modalities added significant complexity Computational cost of high dimensional embedding models Interpretability vs performance	Explainability tools for vector embeddings Less time trying to perfect individual pipelines More time exploring fusion strategies

A series of thin, light brown lines forming an abstract geometric pattern on the left side of the slide. The lines intersect to create various polygonal shapes, some of which are filled with a light beige color.

THANK YOU

Any Questions?